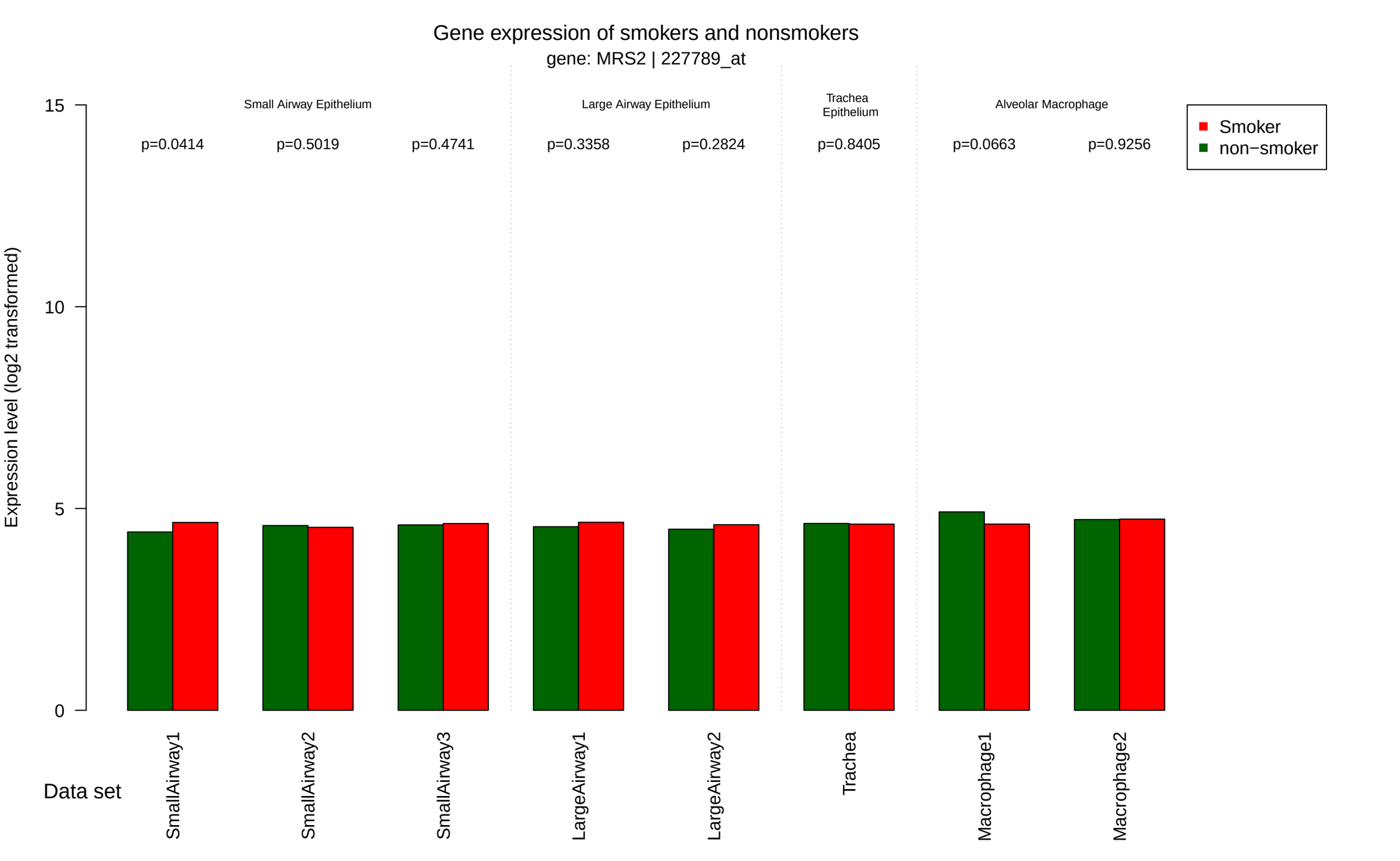
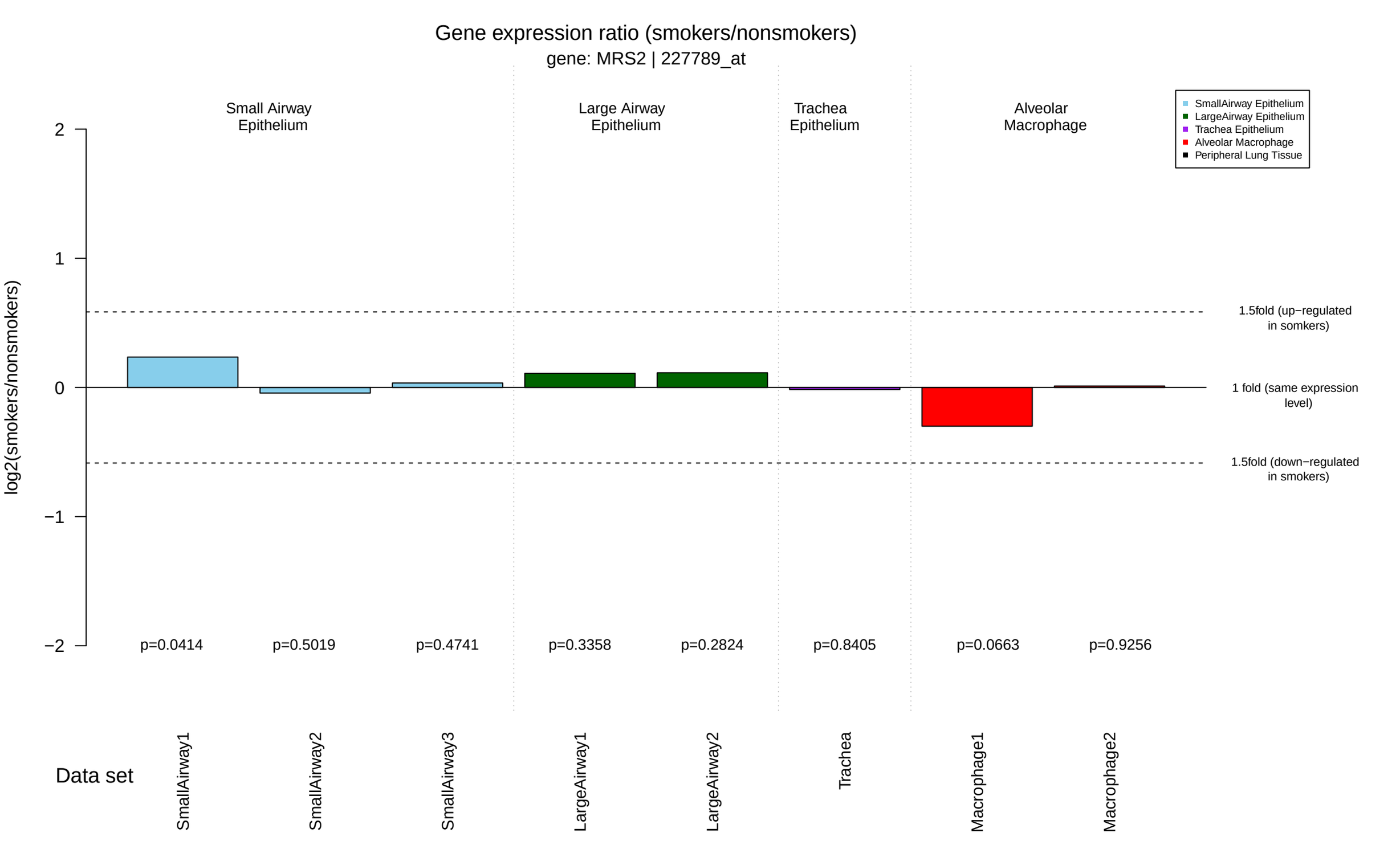
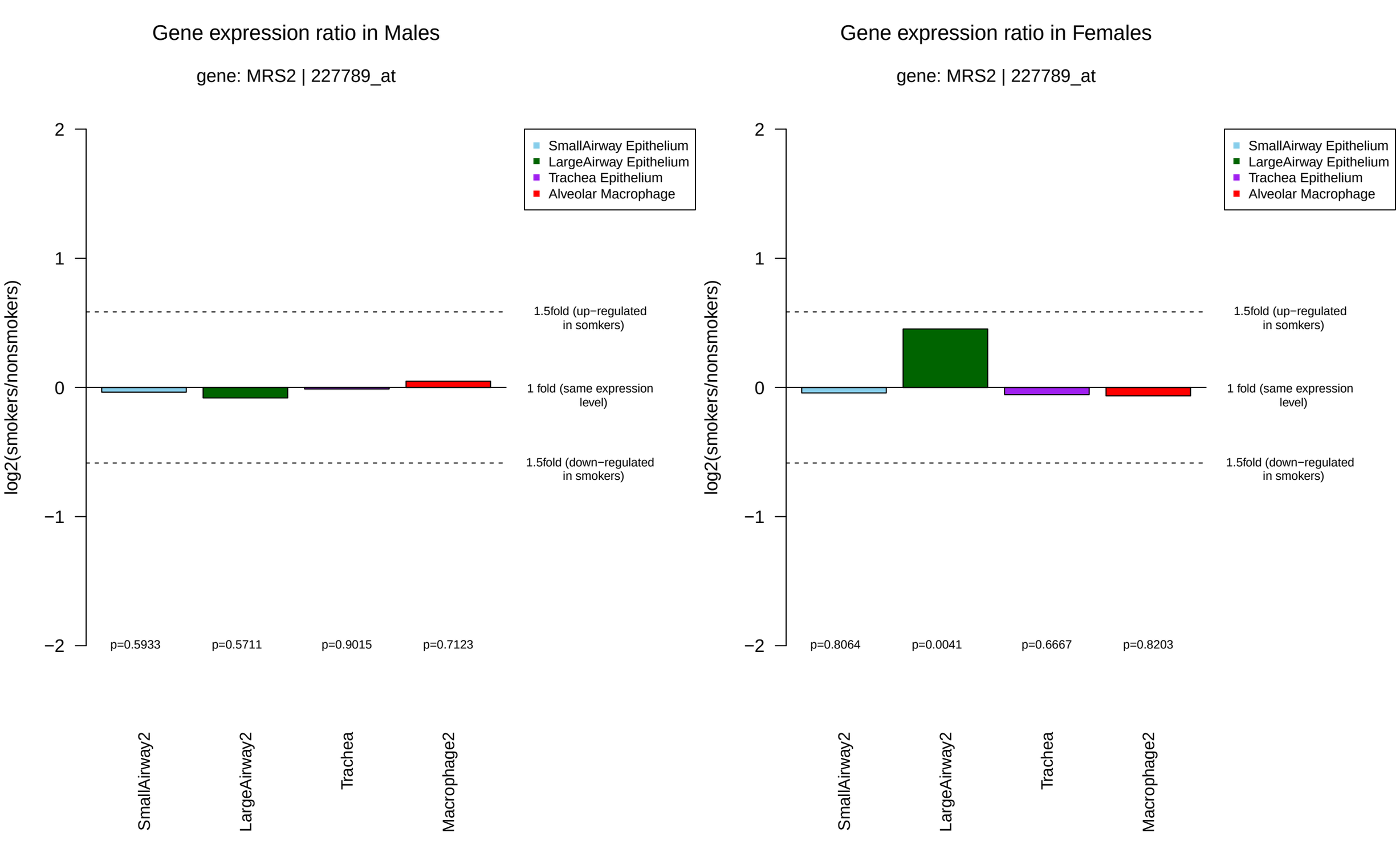
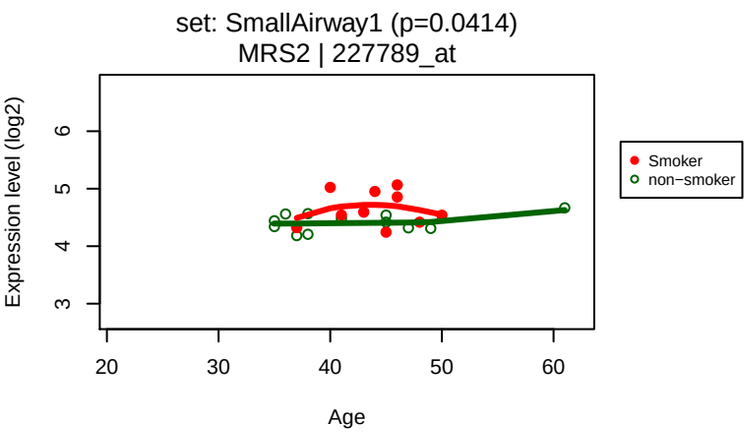
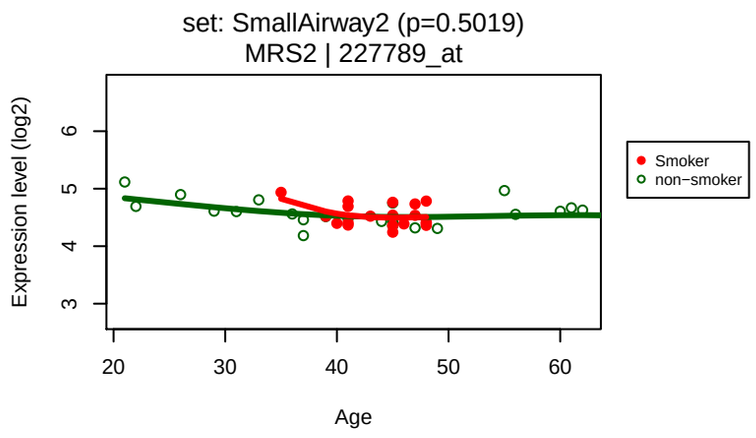
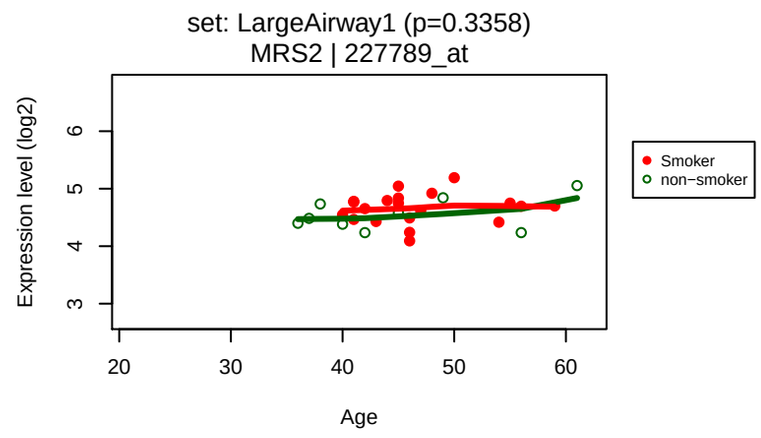

Gene expression of smokers and nonsmokers
gene: MRS2 | 227789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0414
p=0.5019
p=0.4741
p=0.3358
p=0.2824
p=0.8405
p=0.0663
p=0.9256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

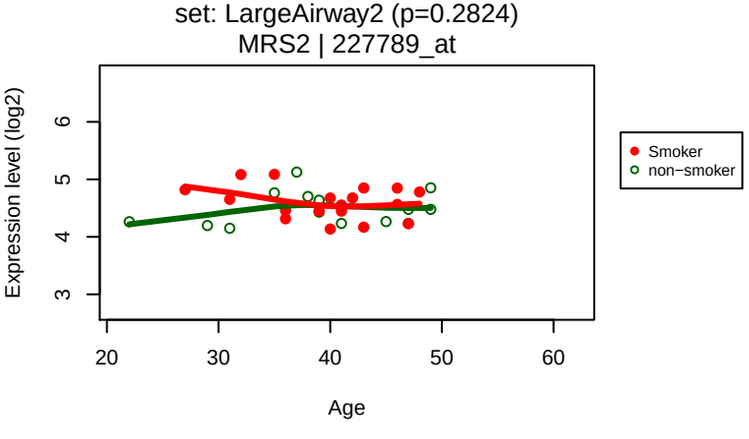

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 227789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0414
p=0.5019
p=0.4741
p=0.3358
p=0.2824
p=0.8405
p=0.0663
p=0.9256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

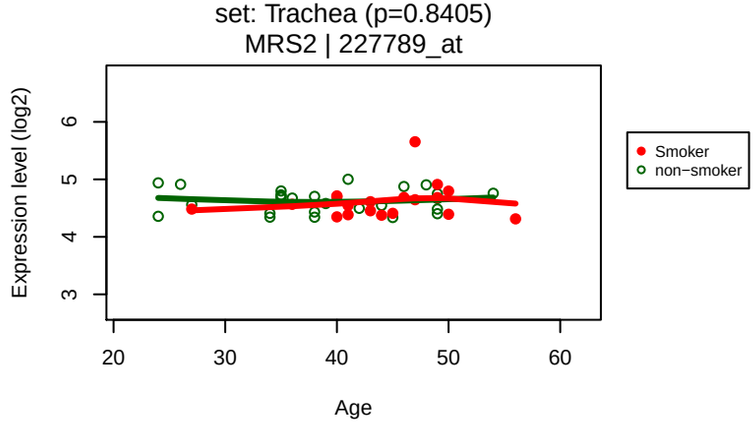

Gene expression ratio in Males
gene: MRS2 | 227789_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5933
p=0.5711
p=0.9015
p=0.7123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 227789_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8064
p=0.0041
p=0.6667
p=0.8203
SmallAirway2
LargeAirway2
Trachea
Macrophage2

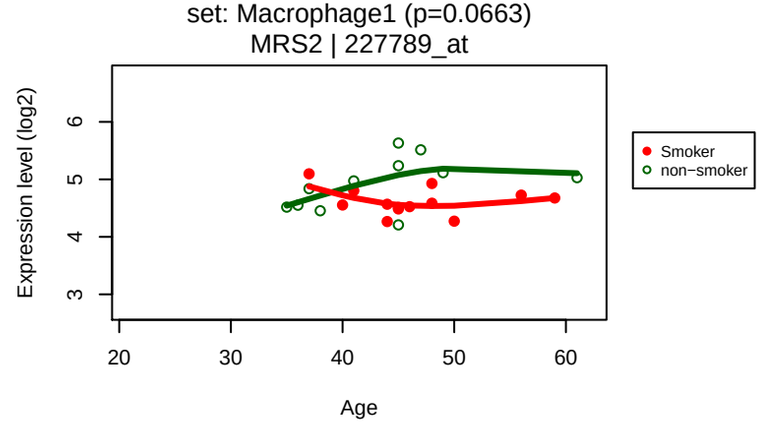

set: SmallAirway1 (p=0.0414)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker

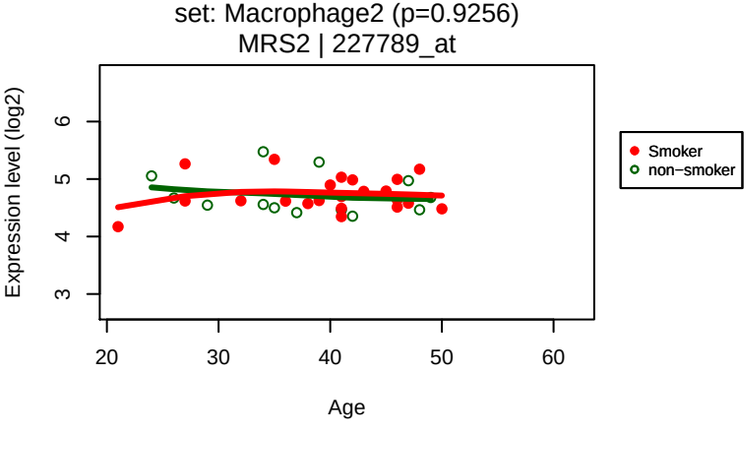

set: SmallAirway2 (p=0.5019)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker

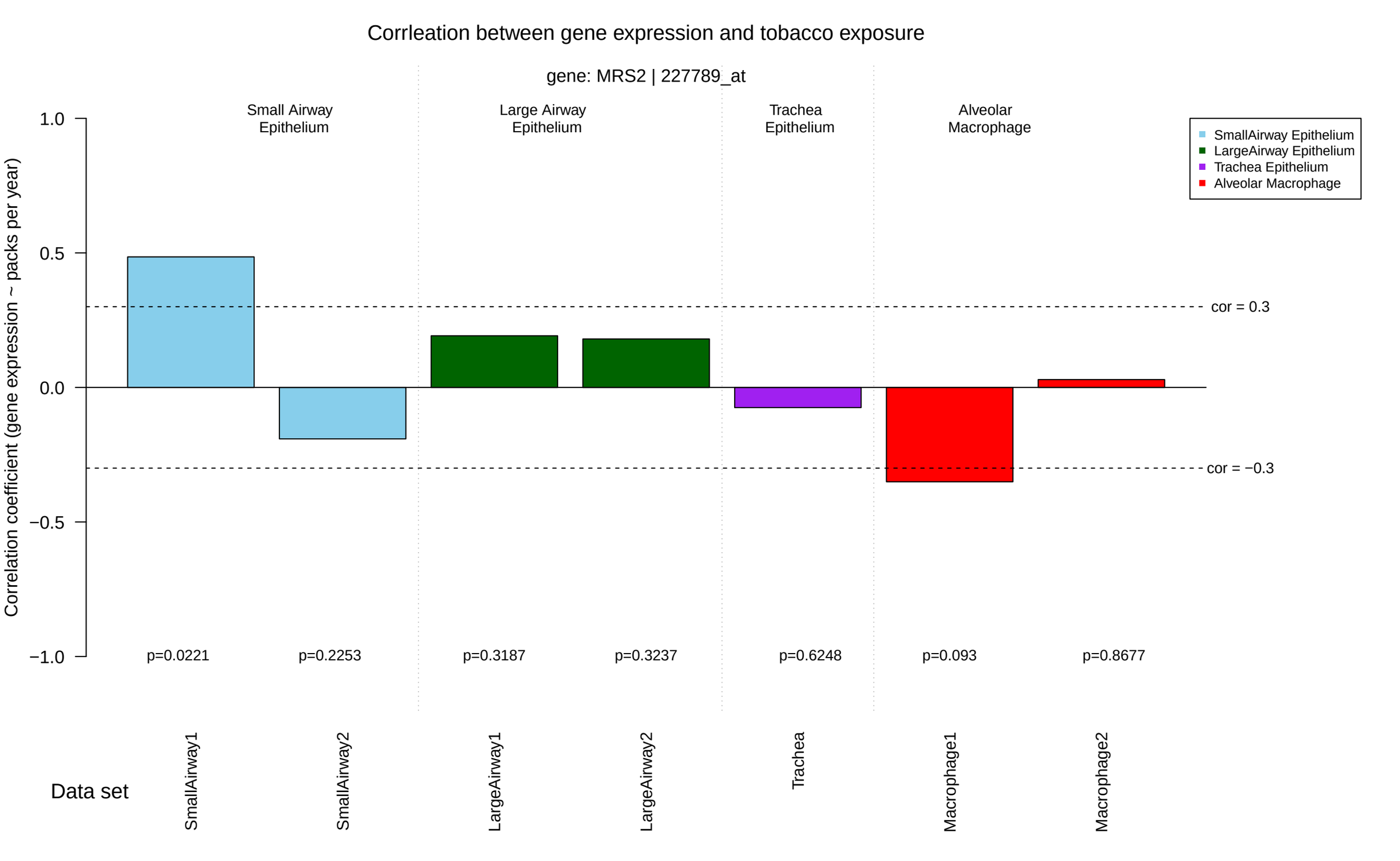

set: LargeAirway1 (p=0.3358)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2824)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8405)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0663)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9256)
MRS2 | 227789_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 227789_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0221
p=0.2253
p=0.3187
p=0.3237
p=0.6248
p=0.093
p=0.8677
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2